Chapter 6
The Topology of Communicating Across Cities of Increasing Sizes, or the Complex Task of "Reaching Out" in Larger Cities

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Abstract Cities have been compared to social reactors constrained by the communication and coordination possibilities offered by an urban environment that has only grown since the advent of the industrial age. We attempt to provide a first description of human interactions in the urban environment using Call Detailed Records (CDR) of the major mobile phone communication network operator in Chile. We build communication networks for 145 Chilean cities to describe and characterize the communication behavior of urban dwellers. We center our analysis in observed indicators of social activity, such as the number of contacts, number of calls and total communication time in each city and evaluate their scaling relationship with the number of mobile phones assigned to each city as an approximation of city size. Interestingly, the values of scaling exponents closely match recent explanations proposed in the literature. The topologies of voice-call networks among cities of increasing sizes are slightly assortative, albeit assortativeness decreases with size. Additionally, they show small average path length relative to their sizes, a typical feature of small-world networks. However, they decrease instead of growing when size is taken into account, unlike other complex networks. Different transitivity indices show mixed results. Average Watts-Strogatz clustering coefficient increases in larger cities much more than expected by pure chance as it has been shown in other social networks. On the other hand, the fact that classic transitivity index decreases seem to exhibit a regime change with a decreasing relation with size and an unexpected growth in larger cities. Both transitivity indices, as a whole, could describe among those who are making new interactions as the city grows. All these results indicate that while tightly knit human communities seem to lose cohesion as they grow, such community properties may progressively disappear among the three to four largest urban centers in Chile where the coordination of complex functions requires each city dweller to reach out to a larger network of people and speak for longer periods

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D. Pumain (ed.), *Theories and Models of Urbanization*, Lecture Notes in Morphogenesis, https://doi.org/10.1007/978-3-030-36656-8_6

of time as compared to smaller cities. Finally, although these results are valid for all networks, there is a division into two regimes when networks reach a critical size of ~10,000 nodes, which raises the possibility of an empirical definition of city for Chile.

6.1 Introduction

The massive amount of data that is usually recorded and stored by telephone companies has high scientific value to unveil how humans interact among each other and with the urban infrastructure. Mobile phone records have been instrumental to revisit studies addressing the intensity and shape of social behavior (Onnela et al. 2007; Candia et al. 2008; Hidalgo and Rodriguez-Sickert 2008; González et al. 2008; Iqbal et al. 2014; Lenormand et al. 2014; Louail et al. 2014; Barbosa et al. 2018), and in particular, social interactivity (Kovanen 2009; Reades et al. 2009; Ratti et al. 2010; Palchykov et al. 2014; Schläpfer et al. 2014). The deep prevalence of mobile phones and smart cards allows to generate contextually detailed descriptions of the networks describing social interactions and their dependence on the structural environment in which they are embedded (González et al. 2008; Hidalgo and Rodriguez-Sickert 2008; Steenbruggen et al. 2014).

While the study of networks is now an active interdisciplinary field, it has historically been linked to the social sciences, where it pioneered the development of metrics and standards to describe the structure of social networks (Jackson 2008; Scott and Carrington 2011). Among the most important metrics used for such purpose are: centrality, a mean to identify the influence of an actor in the network; clustering and transitivity metrics seeking to unveil the presence of microstructures grouping members of the network as a function of various characteristics; and distance or accessibility which seeks to characterize the likelihood of members of the network to find each other (Watts and Strogatz 1998). Later contributions from the field of physics expanded considerably the tools and algorithms used to explore larger datasets while placing particular emphasis on the dynamic patterns responsible for particular topologies.

Important efforts seek to associate topology to function, in order to distinguish empirical networks from random networks with known generative processes that shape their structure (Ben-Naim et al. 2004; Clauset et al. 2009; Newman 2010). The contributions of these research lines to the study of social networks has granted its labeling as a field of its own often referred to as Social Physics (Scott 2011) or Network Science (Barabási and Pósfai 2016; Latora et al. 2017). While this labeling remains a controversial issue, it has sparked an explosion of scientific publications providing important and fresh descriptions of the shape of dynamics, ranging from cellular biology and metabolic chain reactions to particular description of social organization across urban systems (Schläpfer et al. 2014; Zhong et al. 2015).

The large success of the study of networks comes from its ability to provide deep insights based on the description of the internal dynamics of highly interactive systems. Moreover, current data availability and the emergence of new tools have made obvious that any thorough understanding of complex systems is rather incomplete without the effective description of the set of networks linking their individual components (Albert and Barabási 2002; Freeman 2011). The study of networks has thus opened the possibility to explore the dynamic structure of highly interactive systems. It has allowed the generation of specific predictions based on the patterns unveiled by the synoptic descriptions of their topological structure in ecology, epidemiology, economics, transportation, and urban sciences, among others (Estrada 2011).

Various studies suggest that some properties of cities vary systematically with size and may be described by a scaling pattern (Pumain 2004; Bettencourt et al. 2007; West 2017). Empirical analysis show that $Y = Y_0 P^{\beta}$, describes the ubiquitous relation of urban indicators, Y, with P, the size of the city (i.e. urban population), Y_0 a normalization constant and β the scaling exponents, also termed elasticity of Y with respect to P in the economics literature. These regularities, integrate important theoretical developments in biology (Marquet et al. 2005) and statistical mechanics (Barenblatt 1996). The premise is that the interaction among components of complex systems like cities, that emanates from the flows of information, matter, and energy are somehow responsible for the emergence of scaling relationships (Batty 2013; Bettencourt 2013a). However, this approach tends to envision the city as a homogenous system of interacting individuals, which may not be adequate if the city is made of loosely interacting subsystems whose agents (e.g. urban dwellers) take almost independent livelihood decisions. In this context, the integration of urban scaling into more traditional approaches in economics and geography, like those focused on the spatial behavior of individual agents (Thisse 2014), will be key to unveil specific aspects of the underlying dynamics. Moreover, an integration of both views should substantially contribute to develop better theories and tools for a sustainable management of complex urban adaptive systems. For instance, several studies have proposed different models on how interpersonal interactions are structured within a city and how scaling relationships may emerge from them (Onnela et al. 2007; Pan et al. 2013; Martínez 2016; Ribeiro et al. 2017).

While we do not attempt to provide any alternative explanation to scaling, or lack of thereof, we here attempt to contribute to the discussion by providing a thorough description and analysis of the networks of social interaction across cities. We reconstruct and analyze the network of voice call interactions among urban dwellers in Chile. We characterize the topological structure of the network produced from the voice communication between individuals by abstracting the relationships between caller and called and seek to describe changes in the social network topology across urban systems of different sizes. The notion is that of the urban environment impacts on the topological structure of the voice call networks in a country with fairly homogeneous constrains in its socio-demographic and economic composition. This task requires to extend the subject of analysis beyond the particular characteristics of individual cities to, hopefully, unveil macroscopical patterns regarding how social interactions may in fact change with the size of the urban system in which they are embedded.

6.2 Methods

6.2.1 Data

Call Detail Records (CDR) for Chile were provided by Telefonica's research and development center in Chile. Telefonica is among the largest mobile phone providers and holds just under a 40% of the market share across the country. It has a fairly uniform base of users among cities making this dataset well suited for interurban comparisons across the gradient of urban sizes in Chile (Fig. 6.1). We analyzed four weeks of CDR collected during 2015 between March 15th and November 26th summing a total of 5.4×10^8 s of voice conversations (i.e. 171 yrs) among 2,831,085 unique users. This dataset has no information, whatsoever, on the identity of users, nor the content of voice conversations as CDRs were anonymized by the company

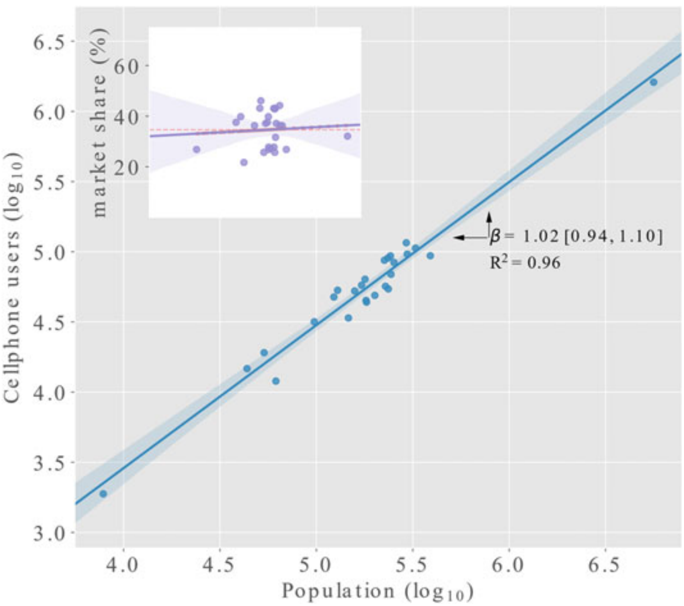

Fig. 6.1 Cell phone users in major urban areas reported by mobile phone provider Telefonica Chile. Market share is calculated as the users in the dataset as a fraction of all users across the different phone companies reported to the governmental telecommunication agency in Chile in 2016 (see: https://www.subtel.gob.cl/estudios-y-estadisticas/telefonia/)

before our analysis. Users ID were available to us as a unique encrypted 32 digit long number. Each record in the CDR contains an identifier for both, the caller and called of a voice interaction. Additionally, the dataset includes the duration, and timestamps for the initiation and termination of the call and the geographic coordinates of the closest antenna where the call was initiated. Since no information on the location of the receiver is available, we used the contemporary Data Detail Record (XDR) dataset to determine the receiver location. XDR are analogous to the CDR but for digital data. A record is produced every time a mobile phone connects to a given antenna to send, or request, a data package. While the number of individual users in the XDR dataset is similar to the CDR, the spatial and temporal density of such records is roughly seven times the size of voice call records and allows to precisely identify the location (i.e. antenna) of the receiver of the call in each CDR by finding the location of the receiver in the XDR right after the call was initiated.

6.2.2 Cities

We used the official definition of cities in Chile to define the geographic extent of urban voice-call interactions. This definition, provided by the Ministry of Housing and Urban Planning,[1] states that "Urbanized areas are bounded by an officially established urban limit with a population larger than 5,000 inhabitants. In the event that a set of spaces with an independent urban boundary are linked together by frequent public transport systems, they are considered as a single city" (Infante Fabres and Pino Castillo 2005). While the definition of urban boundaries may be an important source of heterogeneity in urban scaling analyses (Masucci et al. 2015; Arcaute et al. 2015; Cottineau et al. 2016; Humeres and Samaniego 2017), we use this standard definition as a starting point to evaluate how the communication among urban dwellers change with the size of interaction networks in Chilean cities. Some cities were considered as outliers and removed from the analysis given their erratic behavior leaving the analysis with a total of 145 urban areas in continental Chile. For example, the city of Gorbea (-39.1 lat, -72.7 lon) was removed due to the exceptionally low number of mobile phones ($n = 40$).

6.2.3 Record Selection and Data Pre-processing

To assign call records to urban areas, we approximated the spatial coverage of antennas using Voronoï cells, as done elsewhere (González et al. 2008; Louail et al. 2014). Every call record was assigned to a city if the Voronoï cell of the caller intersected the urban area. Even so, some cells failed to intersect the city in spite of being surrounded

[1] Also: http://observatoriodoc.colabora.minvu.cl/Documentos%20compartidos/OBSERVATORIO%20URBANO%20Listado%20Ciudades_%20Noviembre%202007.xls

by other intersected cells. Such cases where manually assigned to the nearest urban areas. Antennas with no urban assignation were discarded. In order to depict the social interactions within the urban system, we only considered calls in which both, sender and called were located in the same city. This left us with a total of 90,140,299 call records.

6.2.4 Network Construction and Analysis

Undirected networks of voice-call interactions between urban dwellers were built for each city using nodes as users and links between nodes when a voice-call was performed between them. The graph-tool library (ver. 2.19-1) for Python was used both for the construction of networks and its subsequent analysis (Peixoto 2014).

As we were interested in providing a description of how interaction among urban dwellers change across urban systems of different sizes. We first looked at general topological descriptions of networks and later focused on higher order descriptors of group formation among networks for each city (Caldarelli and Vespignani 2007; Newman 2014; Latora et al. 2017). The network aspect described by these indices and the motivation for their usage are described in what follows (see also Table 6.1 for equations).

Ordinary Least Square methods on linearized power equations were used to evaluate scaling relationships between urban network size and network indices. Identification of best models for higher-order topology indices were performed using piecewise regressions. Piecewise, or segmented, regression are used to evaluate the existence of potential thresholds maximizing the fit in a dataset by partitioning the relationship between two or more sections along the x-axis (Hawkins 1976). Akaike Information Criterion (ΔAIC) and the Bayesian Information Criterion (ΔBIC) were used to assess model fit in two segments for all topological attributes (Burnham and Anderson 2003), except for transitivity and density.

6.2.5 Size

The size of networks were computed counting the number of nodes. The size of each urban network is used as a proxy of city size, as the number of urban dwellers is strongly correlated to the number of nodes (i.e. mobile phones users) (Schläpfer et al. 2014).

Table 6.1 Network indices calculated for 147 urban voice-call networks in Chile

Measure	Feature	Notation
Network size	Number of mobile phones	N_i
Number of links	Connections between mobile phones	L_i
Density	Ratio of actual connections	$d_i = \frac{L_i}{N_i(N_i-1)}$
Accumulated degree	Total contacts in city i	$K_i = \sum_n k_n$
Weighted links (calls)	Number of calls ($m \leftrightarrow n$)	$W_i^{(C)} = \sum_m \sum_{n>m} w_{nm}^{(C)}$
Weighted links (time)	Length of calls ($m \leftrightarrow n$)	$W_i^{(T)} = \sum_m \sum_{n>m} w_{nm}^{(T)}$
Number of components	Network structure/fragmentation	c_i
Nodes in largest component	Network structure/fragmentation	S_i
Transitivity index	Probability of common neighbours	$C_i = \frac{3 \cdot \Delta_i}{l(2)_i}$
Mean clustering coefficient	Probability of common neighbours	$\langle C_i' \rangle = \frac{1}{N_i} \sum_n C_{n,i}'$, where $C_{n,i}' = \frac{L^{(n)}}{k_n(k_n-1)}$
Degree assortativity	Similitude between interactings	$r = \frac{\sum jk(e_{jk}-q_j q_k)}{\sigma_q^2}$
Average path length	Social accessibility between non-interactings	$\langle l \rangle = \sum \sum l_{ij}$

6.2.6 *Degree*

The number of links associated to a node was computed as the number of unique voice-calls performed, or received, to other users. This is the fundamental characteristics of any network as the degree of a node (user) is a first indicator of how connected a user is to others in the network. The frequency distribution often serves as a defining characteristic of the different network types (Battiston et al. 2002; Barabási and Pósfai 2016).

6.2.7 *Cumulative Degree*

Computed by counting the number of calls between unique pairs of users and represents the number of voice interactions among different urban dweller per city. Note that the number of links in a network is directly related to the total number of contacts and was measured through the cumulative degree of the network, which is simply twice the number of links (Newman 2014).

6.2.8 Density

Defined as the proportion of links to the total number of potential connections available in a fully connected city network. It usually represents the average connectivity among urban dwellers for a given city.

6.2.9 Component

A component is a group of nodes linked to any other node in the network. Formally, it can be defined using the notion of *path:* a sequence of connected nodes linking any two nodes. Hence, a component is the maximum subset of nodes reachable if we were to follow a particular path. Consequently, the largest component of a network is the component hosting the largest number of available links and is usually called the *major component* (Newman 2014).

6.2.10 Number of Components

The number of components provides a measure of the level of the network's fragmentation. Such fragmentation may be due to the very nature of the network or by the quality of the data gathered to build it. In our case, it provides a mean to identify isolated/segregated groups of users within cities of different sizes.

6.2.11 Fraction of Nodes in the Major Component

With the *number of components*, the fraction of nodes in the major component allows to quantify the fraction of users integrated into the major group of interacting users within the city.

6.2.12 Distance

The distance between two nodes of the network is the minimum number of nodes required to join two nodes along a path. This measure is also known as the "degrees of separation" (Watts and Strogatz 1998).

6.2.13 Average Path Length

That index is the average of the shortest distances between users in the network. It can be interpreted as a measure of the social accessibility among urban dwellers across the city network. We only compute this particular index for the largest component and exclude Santiago, the largest city, due to computational constrains for his calculation (Albert et al. 1999).

6.2.14 Cliques

Cliques are considered as the most basic units of higher order grouping between actors of a network and represents subsets of three interacting nodes formed in a given network. These cliques, or triple bonded nodes, effectively create a triangle of interaction.

6.2.15 Transitivity

Transitivity is commonly used in Social Network Analysis and seeks to quantify a specific characteristic of social groups. In particular, whether *friends of my friends are also my friends*. In formal terms, it represents the transitive relationship between two actors having a common acquaintance in the network (Newman 2014). Usually, partial transitivity in social networks is an indicator that any two interacting individuals have also a social world's similarity (Borgatti and Halgin 2011).

We use two indices to characterize such property, the *Global Clustering Coefficient* initially conceived by Watts and Strogatz (1998) to distinguish empirical networks from random and small world counterparts. This index in defined locally as the fraction of links between neighbors of node n with respect to the total number of links possible to be established. Because this is then averaged over all nodes in the networks (Table 6.1), it suffers from being highly influenced by low-degree nodes (Estrada 2011). Such issue prompts us to consider a second *Transitivity* index defined globally as the rate of cliques to the total number of paths of length two (Newman 2014). Hence, a larger transitivity is indicative of the tendency to belong to a similar social group.

6.2.16 Assortativity

Also seeks to characterize groups within the network, although through a common characteristic. Hence, it represents the preference of actors to be associate themselves to other actors that have a common characteristic such as the income, or the number of acquaintances (Battiston et al. 2002). The most common characteristic used is the degree, for instance. Hence, actors of networks with high degree assortativity prefer to be linked to other actors matching their own degree, meaning that individuals having a large number of contacts will usually communicate with other individuals also having a large number of contacts (Newman 2003).

6.3 Results

The distribution of users among the voice-call networks built for Chilean cities is left skewed with an average of 23,983 users and a median of 4,022. While a large number of components are observed for most urban areas (Fig. 6.2a), the inset of Fig. 6.2b shows that the largest component largely dominates de networks of at least ¾ of the cities. In fact, the fraction of the major component (*S*) rapidly grows to nearly 90% for networks having more than 10,000 users (Fig. 6.2b). At the lower end of the spectrum, we find a small sized mining town, El Salvador, with only 120 (4.9%) isolated subnetworks, and on the upper end, the largest capital city, Santiago, with the largest number of isolated groups of interacting dwellers distributed among 43,459 isolated subnetworks representing only a 4.3% of interacting urban dwellers (Fig. 6.2b).

The frequency distribution of users' degree provides a first view at the shape of voice-call interactions in these urban networks. A comparison of empirical networks with an Erdos-Renyi random network shows that most users exhibit local interactions with a small number of pairs, while others still have a large number of interactions (Fig. 6.2). This produces the large difference of the Erdos-Renyi model at low degrees and the typical heavy tail when using linear binning. However, a closer analysis fitting

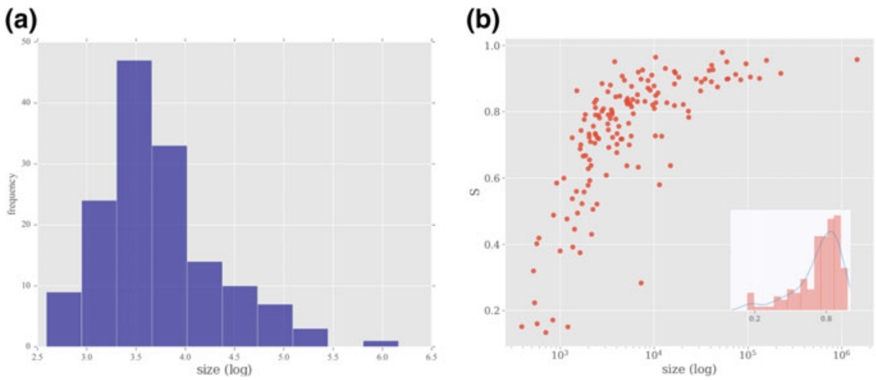

Fig. 6.2 **a** Frequency distribution of the size of voice-call networks in 147 cities in Chile. **b** Relationship between the fraction of nodes in the major components (S) and the size of the voice-call networks among cities in Chile. The inset shows the probability distribution of S

several probability distributions using the method of Clauset et al. (2009) (see also Alstott et al. 2014), suggests that most networks have a scale-free behavior, at least at the tail of their distribution (Fig. 6.3). In fact, fitting lognormal, power law, and a truncated power law distributions favors the latter two for most cities.

Interestingly, no significant relationship seem to exist between the size of the network and the power law exponent α (Pearson correlation: −0.01, p-value: 0.92). However, a larger variation of α is apparent for smaller sized cities compared to larger ones (Fig. 6.4).

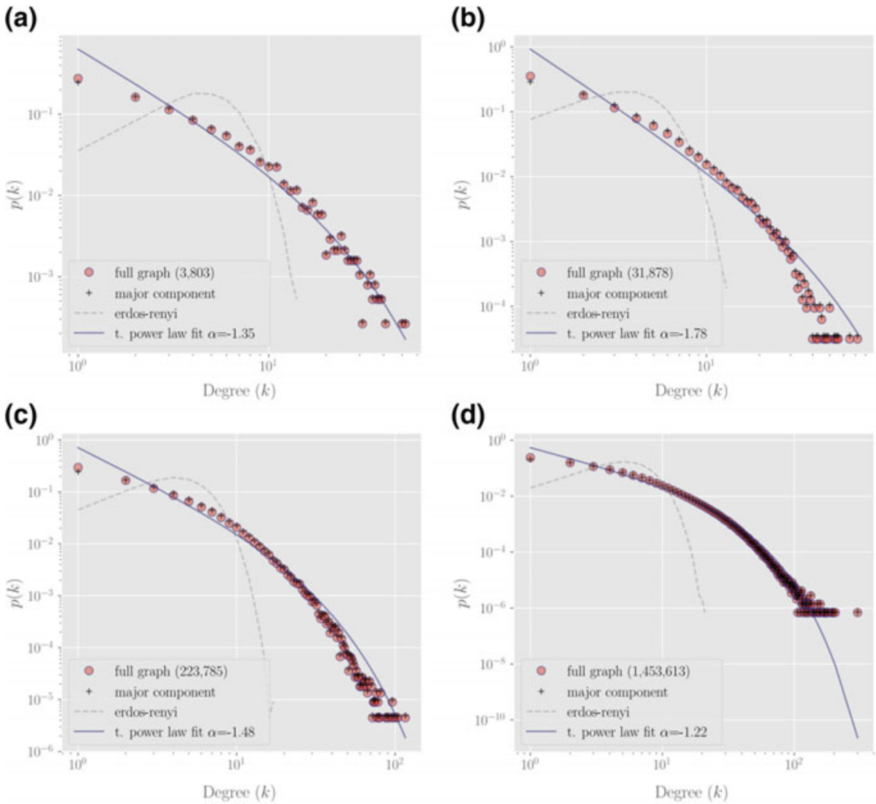

Fig. 6.3 Probability degree distribution for a sample of four cities analyzed in Chile. Cities range from small to large. **a** El Salvador a small mining town of nearly 10,000 people. **b** Valdivia is a medium-to-small university town of nearly 200,000 inhabitants. **c** Gran Valparaíso is the second largest conurbation and includes the city of Viña del Mar, and; **d** Santiago, the largest city in Chile with nearly 5 million urban dwellers. The empirical probability degree distribution is shown for the full graph, the size, i.e. nodes in the network, are in parenthesis. The major component, and a random Erdos-Renyi graph built preserving the original full graph degree distribution. The fit and the exponent of the power law section a truncated power law model is shown in continuous blue line (see text for fitting procedure)

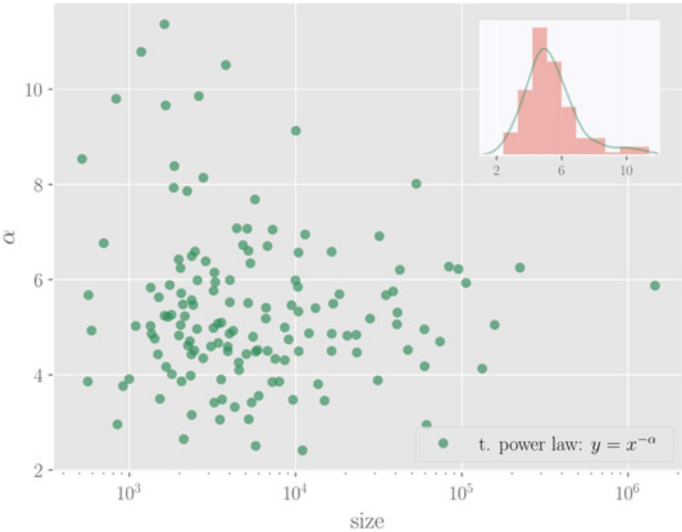

Fig. 6.4 Relationship between size (log) and the exponent α of the fitted truncated power law, for urban voice-call networks in Chile. A truncated power law was fitted to the node degree probability for each network, and the parameter α was then plotted against the size of the network

Further descriptions regarding how urban dwellers establish voice-call interactions may be obtained by inspecting the correlation between nodes. While this may be done for any node characteristics, socio-economic or gender gap for instance, the simplest approach is to correlate average nearest neighbor's degrees, $\overline{k_{nn}}(k)$, or the clustering coefficient, $\overline{c}(k)$, with the degree of each node. Doing this describes a network topology in which Chilean urban system have a slight, albeit very marked correlation between node characteristics at low degrees, k, captured by the exponent of a fitted power law (i.e. α in Fig. 6.5). However, this correlation becomes very heterogeneous as users increase their number of voice-call interactions, k. At such point, the correlation signal fades which, again, suggests the existence of users acting as hubs that favor the interaction with poorly connected pairs. This may be at the origin of the observed scale-free behavior of urban voice-calls as the size of urban networks increases (Fig. 6.5).

The description of topological patterns are complemented by a deeper scrutiny of the interactions among voice-callers in urban Chile. To such end, we looked at an aggregate measure of k, the cumulative degree, K, and its relation to the size of the system where interactions take place (i.e. network size). Doing so, shows that interactions among urban dwellers grow at a rate faster than the size of the network. While this super linear scaling has previously been reported in Schläpfer et al. (2014), the fraction of the total interactions occurring in the network associated to an average user, or density, decreases as the size of cities get smaller (Fig. 6.6). This strongly suggests that, while the size of urban voice-call networks increases, the fraction of interactions per individual gets actually smaller. Hence, while urban

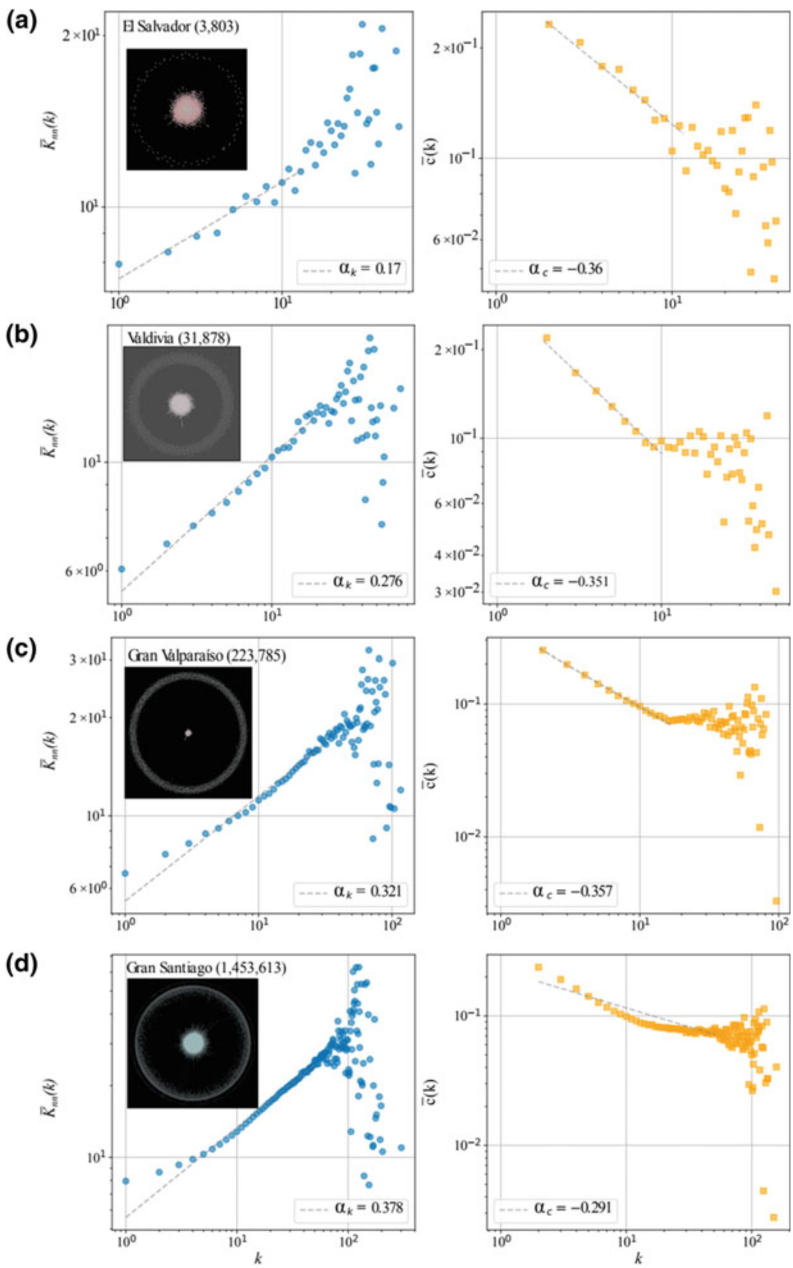

Fig. 6.5 Node degree correlations among four urban voice-call networks (inset). **a** El Salvador (−26.25 lat, −69.63 lon); **b** Valdivia (−39.80 lat, −73.26 lon); **c** Gran Valparaiso (−33.04 lat, −71.62 lon), and **d** Santiago (−33.44 lat, −70.65 lon). The left column shows the correlation between the average degree of node's nearest-neighbors, $\overline{K_{nn}}(k)$, against the degree of nodes, k. The right columns show the correlation between average node clustering, $\overline{c}(k)$ and degree, k. Dotted line is a power law fit with exponent α

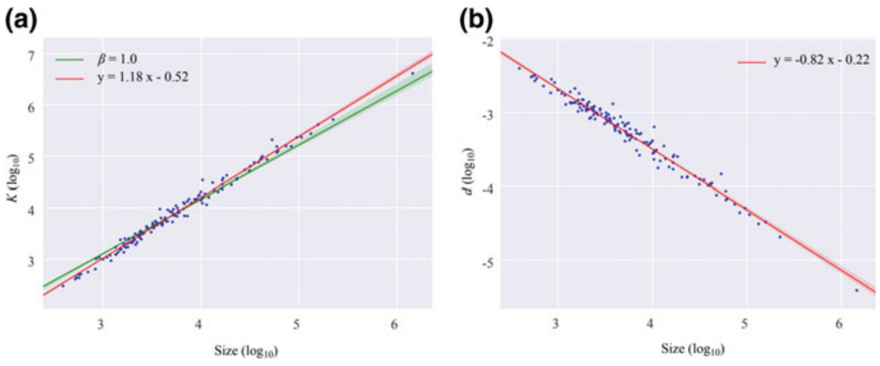

Fig. 6.6 Rates of changes of networks' topological descriptors. **a** There is a superlinear relationship between the cumulative degree and size of the urban voice-call network, meaning that urban dwellers in larger cities tend to have a larger number of voice-call interactions compared to smaller urban areas (linear model fitted using OLS, $R^2 = 0.98$). **b** The rate of change of node density across networks of different sizes suggests that while urban voice-call networks increases, the fraction of interactions per individual decreases, so that the size of the network increases proportionally more that the number of interactions involving an average individual among the Chilean urban system ($R^2 = 0.97$)

dwellers have on average, a larger pool of possible interactions available in larger cities, the number of pairs they actually interact with is a smaller fraction of the whole network. Additionally, the time spent on voice-call interactions and the number of calls (T and W, respectively) also shows a steady, and superlinear, increase with the size of networks (Fig. 6.7). While this might be a mere effect of the size of the network increasing faster than the number of interactions involving an average individual among the Chilean urban system, it also speaks eloquently about how local interaction patterns change among larger cities. Deeper analyses of the robustness of

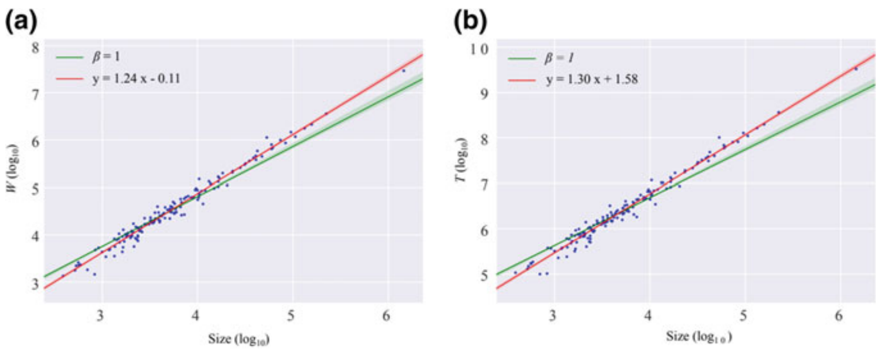

Fig. 6.7 Time and number of communication events across the size of voice-call networks in urban systems in Chile. **a** Shows W, the rate of change in the number of callers as the urban voice-call network change in size ($R^2 = 0.98$). **b** The actual time, T, spent on voice-calls as the size of the urban system gets larger ($R^2 = 0.98$). Both relationship show a superlinear increase

Table 6.2 Estimation of scaling exponent, β, in Fig. 6.6a and 6.7 using a probabilistic framework with explicit error modeling as proposed in Leitão et al. (2016)

	Log-normal error		Normal error		Pearson model
	$\delta = 2$	$\delta \in [1,3]$	$\delta = 1$	$\delta \in [1,2]$	
Cumulative grade (K)	**1.17** ± 0.04	1.17 ± 0.06	1.11 ± 0.09	1.17 ± 0.05	1.10 ± 0.05
Calls (W)	1.22 ± 0.04	**1.23** ± 0.03	1.19 ± 0.08	1.23 ± 0.03	1.18 ± 0.06
Time (T)	1.28 ± 0.04	**1.29** ± 0.02	1.26 ± 0.10	1.29 ± 0.03	1.25 ± 0.06

OLS fits in log-scale is $\delta = 2$. Errors were computed using bootstraps at 95% CI for each model, all models showed p-values <0.05. Grey backgrounds indicate best model for each error types modeling (i.e. log normal, Gaussian, or Pearson). Bold fonts indicate the best overall model. Model selection is based on lowest ΔBIC variation. All estimated parameters show significant differences with a fixed $\beta = 1$. Further details on error modeling procedures are in Leitão et al. (2016)

OLS estimations for such super linearity using various error modeling approaches proposed by Leitão et al. (2016) shows two interesting features. The first is that log-linear regression is the best model for estimating parameters of the scaling of K and size. It also shows that common error modeling using OLS may not hold for W and T, which paradoxically shows no significant differences in the parameter estimation (Table 6.2).

Second order descriptors of urban voice-call networks show how social interactions change with the size of the network. In general terms, transitive properties shown in Fig. 6.8a and b indicates that while networks increase with the size of the city, users have the tendency to interact with pairs having a similar number of interactions, or acquaintances. Now, the joint inspections of both transitivity indices in Fig. 6.8a and b, indicates that the likelihood to be linked to a similarly connected pair increases inversely with the number of links available to users. This is partly due to the fact that the average clustering coefficient (Fig. 6.8a) tends to be biased by low degree nodes compared to the transitivity index proposed by Newman (2003) (Fig. 6.8b) (Estrada 2011). However, the increase in transitivity seems to be more likely associated to a narrowing of the index's variance towards higher transitivity values as the network size increases, reaching a maximum at ~0.12 for all networks (Fig. 6.8a). This is a slightly lower value compared to previously reported values (e.g. in Schläpfer et al. 2014), however, C becomes essentially a constant for urban areas having networks larger than ~10^4 (i.e. green fit in Fig. 6.8b).

While, individuals are more likely to contact other individuals having similar number of contacts (K) for all networks. The magnitude of assortativity decreases as the size of the city increases (p-value $<10^{-3}$) (Fig. 6.8c).

Finally, urban voice-call networks exhibit low average distances, on the order of $log(n)$. This readily suggests themselves as classic *small world* types of networks (Travers and Milgram 1969; Watts and Strogatz 1998).

The variation of higher order network indices across cities of different sizes is documented here, however, the large variability observed in Fig. 6.8 prompts to question the robustness of such patterns across the full range of network sizes. While

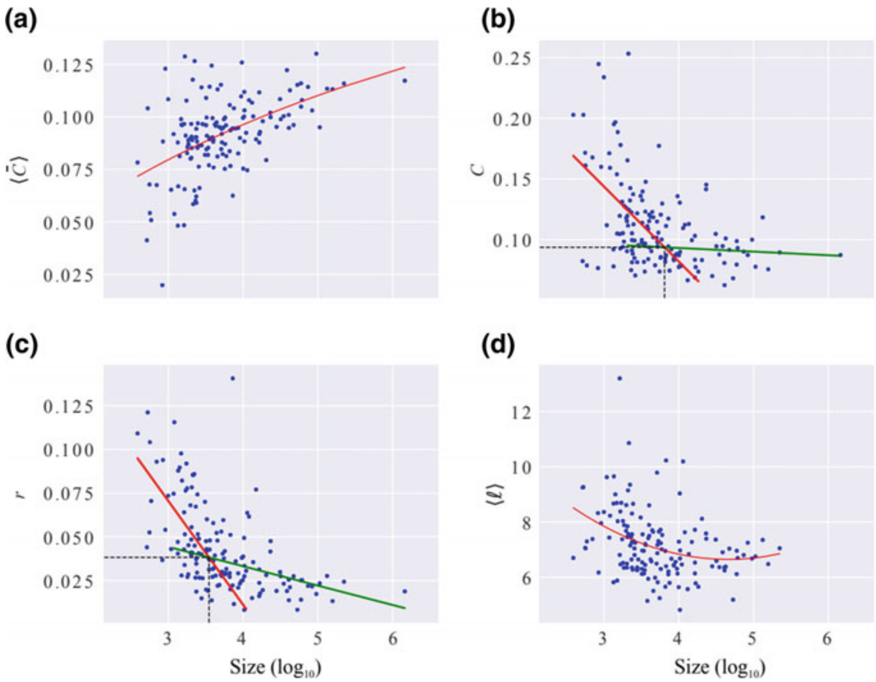

Fig. 6.8 Higher-order topological indices. **a** Average network clustering; **b** transitivity; **c** assortativity; and **d** average path length

it shows that not accounted factors may have important roles in shaping the topology of urban voice-call interaction networks, it may as well points towards the existence of a critical size in which social interactions may operate differently. For instance, a piecewise regression shows the existence of a common threshold when network size reaches ~10^4 (Table 6.1). Thus, our data shows a critical network size that may mark specific shapes in social interactions, however, the large variability observed among small networks could also be a consequence of poor data quality for smaller cities.

6.4 Discussion

While our analysis of urban voice-call networks shows that these networks conforms to topological patterns described for social networks elsewhere (McPherson et al. 2001; Battiston et al. 2002; Newman 2002; Vespignani 2010; Goyal 2011), it is interesting to note that the number of links scale with sizes with a robust statistical regularity (Table 6.2). Hence, while the number of average acquaintances of an urban dweller increases with the size of the city, so does the density. The scaling of the number of links with network size (Fig. 6.6a) shows that sociability increases more

Table 6.3 Parameter estimation of the linear regression of calculated urban voice-call network indices on network sizes

Measure	Linear fit	Range	Confidence interval for β	R^2
Fraction of nodes in largest component	$S = 0.73 \log_{10}(n) - 1.74$	$\log_{10}(n) \leq 3.45$	[0.58, 0.90]	0.62
	$S = 0.09 \log_{10}(n) + 0.44$	$\log_{10}(n) > 3.45$	[0.06, 0.13]	0.22
Transitivity	$C = -0.06 \log_{10}(n) + 0.33$	$\log_{10}(n) \leq 3.81$	[− 0.09, −0.04]	0.19
	$C = 0.00 \log_{10}(n) + 0.11$	$\log_{10}(n) > 3.81$	[− 0.01, 0.01]	0.01
Assortativity	$r = -0.06 \log_{10}(n) + 0.25$	$\log_{10}(n) \leq 3.55$	[− 0.08, −0.03]	0.27
	$r = -0.01 \log_{10}(n) + 0.08$	$\log_{10}(n) > 3.55$	[− 0.02, 0.00]	0.10

For each index, two models were evaluated within the full range of sizes. The best model selected had the lowest variation in Akaike Information Criterion (ΔAIC) and the Bayesian Information Criterion (ΔBIC)

than proportionally, requiring urban dwellers to reach-out to a wider number of pairs as the city increases in size (Bettencourt 2013b; Bettencourt et al. 2014). This pattern has previously been described by Schläpfer et al. (2014), who shows a similar scaling of the cumulative degree (K) with population size, albeit with an exponent, β between 1.05 and 1.13, and a constant transitivity as the size of the network increases. While we show a decrease of transitivity as network size increase, our piecewise regression analysis also show a constant transitivity when using Newman (2003) global index for cities larger than ~$10^{3.8}$ users (Table 6.3), suggesting that interaction dynamics may shape differently for very small networks.

The seemingly opposed trends shown by the transitivity metrics employed may be indicative of how this property changes with the network size. The fact that average clustering coefficient is strongly influenced by low-degree nodes (Newman 2003), points to the existence of two phenomena as the size of the voice-call network increases. On one hand, there seem to be an increase in the proportion of poorly connected users forming clusters of low-degree individuals; and, on the other hand, as the size of networks increases, the number of highly connected individuals tend to interact with poorly connected pairs, forming *hubs* acting as bridges that connects communities of low-degree individuals. Such heterogeneity apparently typical of large social networks was extensively studied in social physics (Borgatti and Everett 2000) and similar patterns were recently described by Onnela et al. (2007). Recently various mechanisms have been proposed to explain the emergence of this type of networks such as the *preferential attachment* model in which new users fill prefer to interact with highly connected pairs (Barabási and Albert 1999) or the *rich-club* *effect* model describing how highly connected users will mostly establish connections among each other (Opsahl et al. 2008; Leo et al. 2016).

The drop of assortativity with cities size reflects that contact preferences between individuals of similar social activity is reduced as the size of the urban environment grows. This may be associated to the well described constraints imposed by larger, more heterogeneous, and complex organization of cities through congestion (Louf and Barthelemy 2014; Barthelemy 2016), or structural constraints of urban infrastructure on urban life (Samaniego and Moses 2008; Louf and Barthelemy 2014). In fact, while social and economic diversity has largely been linked to more productive urban systems (Lobo et al. 2013; Bettencourt et al. 2014; Youn et al. 2016), heterogeneity among urban dwellers is syndicated as a major determinant of network structure by sociologists (i.e. regular equivalence index; Hanneman and Riddle 2011). Such observation points towards portraying larger cities as effectively more unequal or concentrated in terms of their social relations, as they become more diverse in terms of the structures of each individual's social networks.

6.5 Conclusion

We here present a topological analysis of social networks derived from telephone call records (CDR). Using standard metrics, we have described the properties of social networks in cities, and observed how these evolve with the size of the city. The majority of our findings coincide to a large extent with empirical observations and theoretical models related to the greater inequality and diversity of social interactions present in larger cities. Even when larger cities have less social accessibility compared to small cities, the results show that in general Chilean cities continue to have *small world* behavior.

This work, and others, clearly suggest that cellphone connections may be a useful proxy of real social interaction (Coscia and Hausmann 2015). This is embedded in the well-known discussion seeking to understand the relationship between social interaction and urban infrastructure (Picornell et al. 2015; Alessandretti et al. 2018; Carrasco et al. 2018). As Carrasco et al. (2008) clearly puts it, this dependence is clearly justified by recognizing that a large part of individual's motivation relates to "*…with whom they interact rather than where they go*".

We hope that our interpretations and conclusions will contribute to the understanding of human interactions in the urban environment. Particularly to foster new approaches and usage of the vast amount of digital traces generated to understand urban segregation and inequality.

Acknowledgements We thank Pablo García and the team at Telefónica I+D Chile for the provision of anonymized X/CDR data for Chile.

Funding Funding for this research was provided by the Chilean research council CONICYT under the grant program FONDECYT #1161280 to Horacio Samaniego.

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